L18_Lecture Scribe

Generate a lecture scribe suitable for exam preparation. While writing the scribe, reason step by step through definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce material that is not present in the context.

Below is a **structured lecture scribe for exam preparation** based strictly on the slides you provided. I preserved the **logical order of definitions → derivations → properties → examples**, and explained the **reasoning step-by-step exactly following the lecture flow** without introducing external material.

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

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1. Marginal PMF

1.1 Definition

Consider two **discrete random variables** X and Y.

The **joint PMF** is defined as:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

This gives the probability that **both events occur simultaneously**.

However, often we want the probability distribution of **only one variable**.

To obtain it, we **sum the joint probabilities over all possible values of the other variable**.

Thus:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from a Joint PMF Table

Suppose the joint PMF is given in a table.

$p_{XY}(x, y)$	y_1	y_2	y_3	$p_X(x)$
x_1	p_{11}	p_{12}	p_{13}	$\sum_j p_{1j}$

$p_{XY}(x,y)$	y_1	y_2	y_3	$p_X(x)$
x_2	p_{21}	p_{22}	p_{23}	$\sum_j p_{2j}$
x_3	p_{31}	p_{32}	p_{33}	$\sum_j p_{3j}$

From this table:

$$p_X(x_i) = \sum_j p_{ij}$$

Similarly:

$$p_Y(y_j) = \sum_i p_{ij}$$

Important property:

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

Thus the marginals are valid probability distributions.

1.3 Example: Computing Marginal PMFs

Suppose the joint PMF is

$p_{XY}(x,y)$	$y=0$	$y=1$
$x=0$	$1/8$	$1/4$
$x=1$	$3/8$	$1/4$

Step 1: Compute $p_X(x)$

For $x=0$:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

For $x=1$:

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Thus

$$p_X(x) = \begin{cases} 3/8 & x=0 \\ 5/8 & x=1 \end{cases}$$

Step 2: Compute $p_Y(y)$

For $y=0$:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{4}{8} = \frac{1}{2}$$

For $y=1$:

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Thus

$$p_Y(y) = \begin{cases} 1/2 & y=0 \\ 1/2 & y=1 \end{cases}$$

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2. Correlation

2.1 Definition

A basic measure of the **joint behavior** of two variables is the **product moment**:

$$R_{XY} = E[XY]$$

For continuous random variables:

$$R_{XY} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y)\, dx\, dy$$

This represents the **average value of the product** XY.

It is often called a **raw correlation measure**.

2.2 Limitations

The raw product moment has some issues:

- It **depends on the scale of** X **and** Y.
- It is **computed about the origin**, not about the mean.

Therefore it mixes together:

- mean values
- spread
- dependence

Thus it is **not a clean measure of centered dependence**.

To remove the effect of means, we **center the variables**.

This leads to the concept of **covariance**.

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3. Covariance

3.1 Definition

Covariance measures how two variables vary **together around their means**.

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation:

1. First subtract the mean from each variable.
2. Then measure how their **centered deviations vary together**.

Thus covariance captures whether deviations occur in the **same direction or opposite directions**.

3.2 Interpretation

Positive Covariance

$$\mathrm{Cov}(X, Y) > 0$$

When one variable increases, the other tends to increase.

Example: **GPA vs hours studied**

Negative Covariance

$$\mathrm{Cov}(X, Y) < 0$$

When one increases, the other tends to decrease.

Example: **speed vs travel time**

Zero Covariance

$$\mathrm{Cov}(X, Y) = 0$$

No linear co-variation trend.

Important note:

Zero covariance means **uncorrelated**, but **does not necessarily imply independence**.

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4. Algebraic Form of Covariance

Start with the definition

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the product:

$$(X - \mu_X)(Y - \mu_Y) = XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y$$

Taking expectation:

$$E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Using linearity of expectation:

$$E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since

$$E[X] = \mu_X, \quad E[Y] = \mu_Y$$

Substitute:

$$E[XY] - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y$$

Simplifying:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

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5. Properties of Covariance

1. Symmetry

$$\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$$

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2. Variance as a Special Case

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X)$$

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3. Linearity with Constants

$$\mathrm{Cov}(aX + b, cY + d) = ac\,\mathrm{Cov}(X, Y)$$

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4. Independence

If

$$X \perp Y$$

then

$$\mathrm{Cov}(X, Y) = 0$$

But the reverse is **not always true**.

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6. Example 1

Given

$$Y = -2X + 3$$

Find

$$\mathrm{Cov}(X, Y)$$

Step 1: Use covariance identity

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Compute XY

$$XY = X(-2X + 3)$$
$$XY = -2X^2 + 3X$$

Thus

$$E[XY] = E[-2X^2 + 3X]$$

Step 3: Compute $E[Y]$

$$Y = -2X + 3$$

So

$$E[Y] = -2E[X] + 3$$

Step 4: Substitute

$$\mathrm{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Simplifying gives

$$\mathrm{Cov}(X, Y) = -2E[X^2] + 2(E[X])^2$$

But

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Thus

$$\mathrm{Cov}(X, Y) = -2\mathrm{Var}(X)$$

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7. Pearson Correlation Coefficient

Covariance depends on units.

To obtain a **normalized measure**, we divide by standard deviations.

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

Properties:

- Measures **strength and direction** of linear relationship
- **Unit-free**
- Normalized version of covariance

Range:

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation:

- $\rho > 0$ → positive linear trend
- $\rho < 0$ → negative linear trend
- $\rho = 0$ → no linear correlation

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8. Example 2 (Joint PDF)

Given joint density

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0 & \text{otherwise} \end{cases}$$

Compute:

1. $E[X]$
2. $E[Y]$
3. $E[XY]$
4. $E[2X + 3Y]$
5. $Var(X)$
6. $Var(Y)$
7. $Cov(X,Y)$
8. ρ_{XY}

Step 1: Compute $E[X]$

$$E[X] = \int_0^4 \int_1^5 x f_{XY}(x,y)\,dy\,dx$$

Substitute the PDF:

$$E[X] = \int_0^4 \int_1^5 x \frac{xy}{96}\,dy\,dx$$

Evaluating the integrals gives

$$E[X] = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$E[Y] = \int_1^5 y f_Y(y)\,dy$$

Result:

$$E[Y] = \frac{31}{9}$$

Step 3: Compute $E[XY]$

Since X and Y are independent,

$$\begin{aligned} E[XY] &= E[X]E[Y] \\ &= \frac{8}{3} \times \frac{31}{9} \\ &= \frac{248}{27} \end{aligned}$$

Step 4: Expectation of Linear Combination

Using linearity:

$$\begin{aligned} E[2X + 3Y] &= 2E[X] + 3E[Y] \\ &= 2\left(\frac{8}{3}\right) + 3\left(\frac{31}{9}\right) \end{aligned}$$

$$= \frac{47}{3}$$

Step 5: Variance of X

$$Var(X) = E[X^2] - (E[X])^2$$

Calculation yields

$$Var(X) = \frac{8}{9}$$

Step 6: Variance of Y

$$Var(Y) = E[Y^2] - (E[Y])^2$$

Result:

$$Var(Y) = \frac{92}{81}$$

Step 7: Covariance

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$
$$= \frac{248}{27} - \frac{248}{27}$$
$$= 0$$

Thus

$$Cov(X, Y) = 0$$

Step 8: Correlation

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Since $Cov(X, Y) = 0$

$$\rho_{XY} = 0$$

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9. Jointly Gaussian Random Variables

Definition

A pair (X, Y) is **jointly Gaussian** if:

- Their **joint density is bivariate Gaussian**
- Each marginal is Gaussian.

$$X \sim N(\mu_X, \sigma_X^2)$$
$$Y \sim N(\mu_Y, \sigma_Y^2)$$

Joint Gaussian PDF

The joint density is

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where

- μ_X, μ_Y = means
- σ_X, σ_Y = standard deviations

- ρ = correlation coefficient

Real-life Example

Example application:

Climate system variables:

- X: atmospheric CO_2 concentration (ppm)
- Y: global surface temperature anomaly (°C)

Environmental processes introduce randomness, so their **joint behavior can be approximated by a joint Gaussian model**.

If

$$\rho > 0$$

then **higher CO_2 levels tend to occur with higher temperatures**.

✅ This completes the **full lecture scribe for revision** following the **exact lecture flow**.

If you want, I can also convert this into a **10–12 page exam-ready PDF style scribe (with clearer derivations and boxed formulas for memorization)** which usually makes **Probability exams much easier to revise.**